1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that immune or molecular signals (e.g., immune infiltrates, MSH6 status, EV mRNA signatures) do not always translate into improved outcomes — i.e., why do some immune-infiltrated MIBC tumors still have poor prognosis? 2) Tier 1 paths used: I built on MSH6 (canonical MMR gene whose alteration shapes mutational/neoantigen landscape) and pZVR (reported independent predictor of lymph node metastasis/metastatic propensity). These provide a direct mechanistic axis linking tumor genomic instability to metastatic behavior that can subvert immune benefit. 3) Tier 2 support: EV mRNA markers, KRT17 and MDK (recurrence/metastasis signals) support a non-invasive readout of aggressive biology; MSI/Immunoscore and AI-based biomarker paths justify immune-contexture assessment and scalable detection approaches that can stratify the proposed subgroups. 4) Endpoint choice: Overall survival (OS) was selected because the synthesis focused on prognosis and nodal/metastatic predictors (pZVR) and because OS is the most definitive clinical outcome for assessing whether immune signals translate into durable benefit. 5) Deliberate exclusions: I excluded EV-mRNA markers and MDK/KRT17 from the core claim to keep the mechanistic hypothesis sharp (max 2 anchors); these are placed as secondary_modifiers to be tested as mediators or readouts. HDL-C, PTGDS and exploratory CSF/PRAD links were omitted from the core hypothesis because they are either systemic/metabolic or cross-cancer analogies with weaker bladder-specific mechanistic support?

Working Hypothesis

Primary hypothesis: In muscle-invasive bladder cancer (MIBC), the survival benefit associated with high tumor-infiltrating lymphocyte (TIL) density is conditioned by MSH6 status. Specifically, in MSH6-proficient tumors, high TIL density predicts improved overall survival (OS). In contrast, in MSH6-deficient tumors with mismatch repair dysfunction, intratumoral neoantigen heterogeneity and clonal diversity promote metastasis-prone subclones, thereby negating or reversing the TIL-associated survival advantage and leading to equal or worse OS compared with TIL-low disease. This interaction is further modulated by Tier2 modifiers—KRT17, MDK, MSI status, Immunoscore-like metrics, and extracellular vesicle (EV) mRNA readouts—that reflect immune contexture and aggressive tumor biology and can subdivide patients into distinct risk groups. The primary endpoint is OS; secondary endpoints include disease-free survival (DFS), metastasis-free survival (MFS), and response to adjuvant therapy. The pZVR biomarker is incorporated as a surrogate predictor of metastatic propensity within the modeling framework. HDL-C and PTGDS are not posited as core drivers of the interaction but may be explored as ancillary systemic factors. Testing plan involves multi-omic integration and prospective validation to determine whether TIL-high/MSH6 status defines a niche where alternative or intensified adjuvant strategies are warranted.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, pZVR
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 (canonical MMR gene whose alteration shapes mutational/neoantigen landscape) and pZVR (reported independent predictor of lymph node metastasis/metastatic propensity).

pZVR: pZVR) and because OS is the most definitive clinical outcome for assessing whether immune signals translate into durable benefit.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	EV mRNA markers	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	MDK	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	Immunoscore-like metrics	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective cohorts with comprehensive pathology (TIL density, MSH6 IHC/ sequencing) and multi-omic data (WES for mutational burden and clonal architecture, neoantigen inference, and Tier2 marker assays);

Prediction 2: digital pathology and AI-assisted scoring for reproducible TIL density and Immunoscore-like metrics;

Prediction 3: multi-region or single-cell sequencing to quantify intratumoral heterogeneity and clonal dynamics;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EV mRNA markers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- KRT17: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.